



HL7 AI CHALLENGE 2026 · EFFICIENCY & EFFECTIVENESS

Cross-Border IPS AI Agent

Federated terminology learning + FHIR-native mappings + an International Patient Summary (IPS) Bundle

Local clinical language becomes coded, auditable, and exchangeable across borders.

TEAM · TECH ADAPTIVE PANDIT

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IPS AI Agent
Federated terminology + FHIR R4 IPS-style document Bundles

4 sites
IPS 2.0.1: 0 errors

step 01

down-up (default)

Target
IND

Summary / source PDF
human
ing document. HL7 FHIR m-facing interoperability
d USA source PDF

pe 2 diabetes
year old male with
ood pressure has
creatinine 1.1
metformin 1000 mg
.0 mg.
it used by the validated

un pipeline

try
the local site.
ses stay local.
ay local.
undles stay local.
s model tensors + sample

CURRENT EXCHANGE ROUTE
United States > India
semantic interoperability
FHIR document exchange
ABDM I

WHY HL7 MATTERS HERE
Country PDFs are readable documents; HL7 FHIR IPS is the interoperable source of truth.
The demo converts local clinical wording into standard codes, packages them in FHIR IPS, then renders a receiver-friendly country report from that Bundle.
HL7-centered exchange
SOURCE COUNTRY PDF
US Patient Care Summary / C-CDA-style source PDF
USA human-readable clinical input
HL7 INTEROPERABILITY LAYER
HL7 FHIR IPS document Bundle
canonical machine-readable source of truth for exchange
TARGET COUNTRY PDF
Final Indian receiver
ABDM FHIR R4 readiness via rendering

CASE MAPPING COVERAGE
100%
6/6 facts · registry + FL
FHIR BUNDLE TYPE
document
IPS-style FHIR R4
OFFICIAL IPS VALIDATION
0 errors
0 warnings · 4/4 Bundles
SYNTHETIC
48 / 4
48 received

01 Report > 02 Facts > 03 Terminology > 04 FL Linker > 05 ConceptMap > 06 IPS Bundle >

End-to-end clinical QA path
What makes the Bundle medically reviewable, not just structurally valid.
SOURCE-TO-FACT
6 retained facts
source phrases stay visible for audit
FACT-TO-CODE
SNOMED CT + LOINC + RxNorm
+ ICD-10
registry first, federated linker on misses
CODE-TO-FHIR
Condition / Observation / MedicationStatement
clinical facts placed into expected resources
DOCUMENT P
Composition first
Bundle.type=document id

Clinical Trace
FHIR Terminology
Federated Learning
FHIR IPS Bundle
Evidence & Limitations

Clinical fact extraction → terminology binding
Rule-backed extractor (prototype). Pretrained clinical NER is future work.
6 facts
SNOMED CT · ICD-10

PHRASE	NORMALIZED CONCEPT	CODE(S)	FHIR RESOURCE
T2DM	Type 2 diabetes mellitus	SNOMED CT 44054006 ICD-10 E11	Condition

Make local clinical language computable across borders

PROBLEM

Receiver EHRs cannot interpret local terms

OBJECTIVE

One coded, machine-readable patient summary

SOLUTION

Federated terminology learning + FHIR IPS

IMPACT

Auditable sharing; less manual recoding

**CORE THESIS**

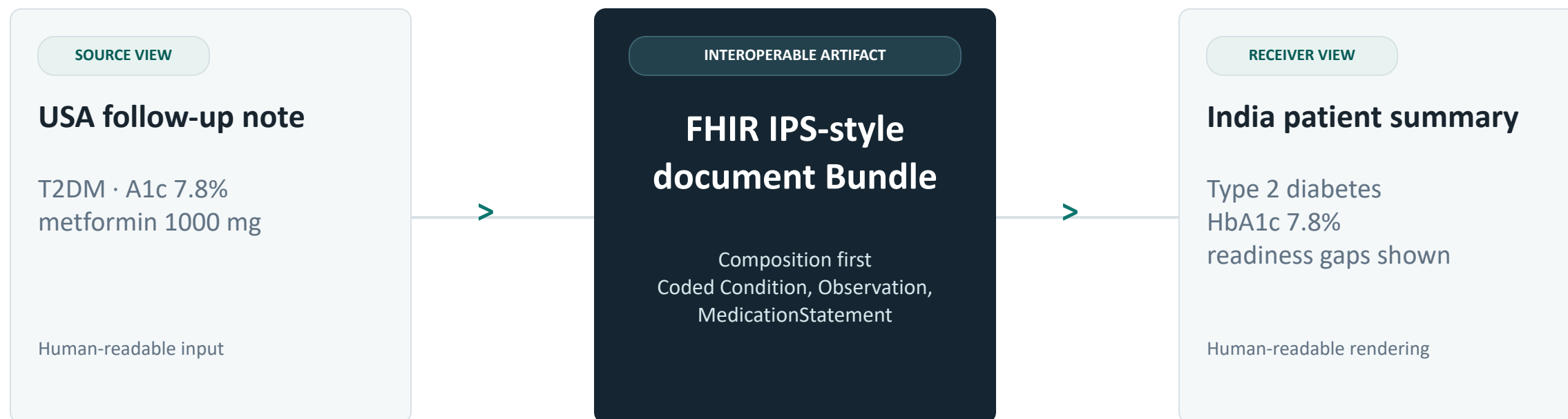
The interoperable summary travels.

The learning stays local.

FHIR terminology + IPS

The deck proves this with a synthetic benchmark, FHIR-native mapping traces, and official IPS validator evidence.

One coded summary supports a cross-border handoff

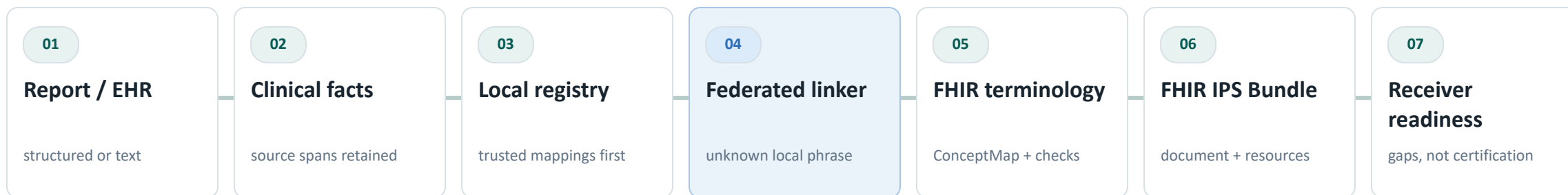


FHIR IPS is the machine-readable cross-border artifact; PDFs remain human-readable views.

Expected impact: less recoding, traceable handoffs

Every mapping is traceable from source phrase to FHIR

Registry first. Federated linker for unknown phrases. FHIR terminology checks. Then IPS packaging.



AUDIT TRACE

SOURCE PHRASE

T2DM

CANONICAL CONCEPT

> **Type 2 diabetes mellitus**

STANDARD CODES

> **SNOMED CT 44054006 · ICD-10 E11**

FHIR DESTINATION

> **Condition resource**

Low-confidence predictions stop for human review and do not create trusted clinical codes.

Federated learning aligns local terms without pooling reports

Across five seeds, the federated model mapped all 48 transfer phrases; local-only models missed one.

768

local training phrase examples

47/48

local-only correct transfer mappings

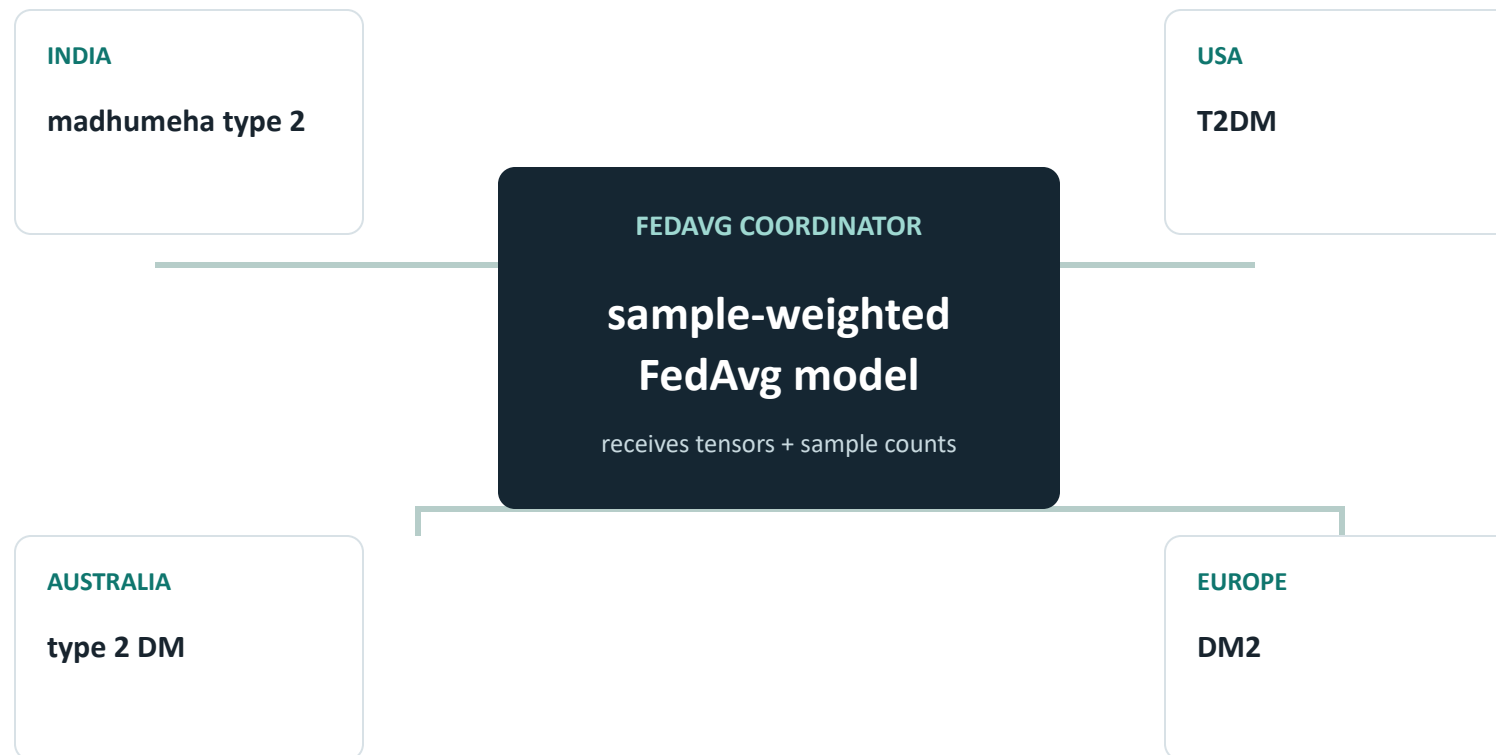
48/48

federated correct transfer mappings

192/192

globally unseen test / seed · macro-F1 1.000

48.05 KiB/client update · 1.88 MiB total / 5 rounds



Data locality, not cryptography: model updates may leak information without secure aggregation or differential privacy.

The mapping is an auditable HL7 artifact, not a hidden AI guess

MAPPING TRACE

SOURCE	India local term madhumeha type 2
PREDICTION	Type 2 diabetes mellitus confidence 0.94
CONCEPTMAP \$TRANSLATE	SNOMED CT 44054006 equivalent
\$LOOKUP	target code exists display resolved
\$VALIDATE-CODE	member of allowed condition ValueSet
RESULT	trusted mapping reusable local cache entry

confidence < 0.70 -> human review; no code emitted

Local CodeSystem

Represents the site's local phrase or code.

Target ValueSet

Constrains candidate concepts by clinical category.

FHIR ConceptMap

Publishes source-to-target mapping with provenance.

Terminology operations

\$translate maps; \$lookup resolves; \$validate-code checks.

A valid target code does not prove the semantic mapping is correct. Gold labels, trace review, and clinician adjudication remain required.

All four checked cross-border Bundles passed with zero errors

4/4

Validated against FHIR R4 + IPS 2.0.1

0

errors

0

warnings

Composition first

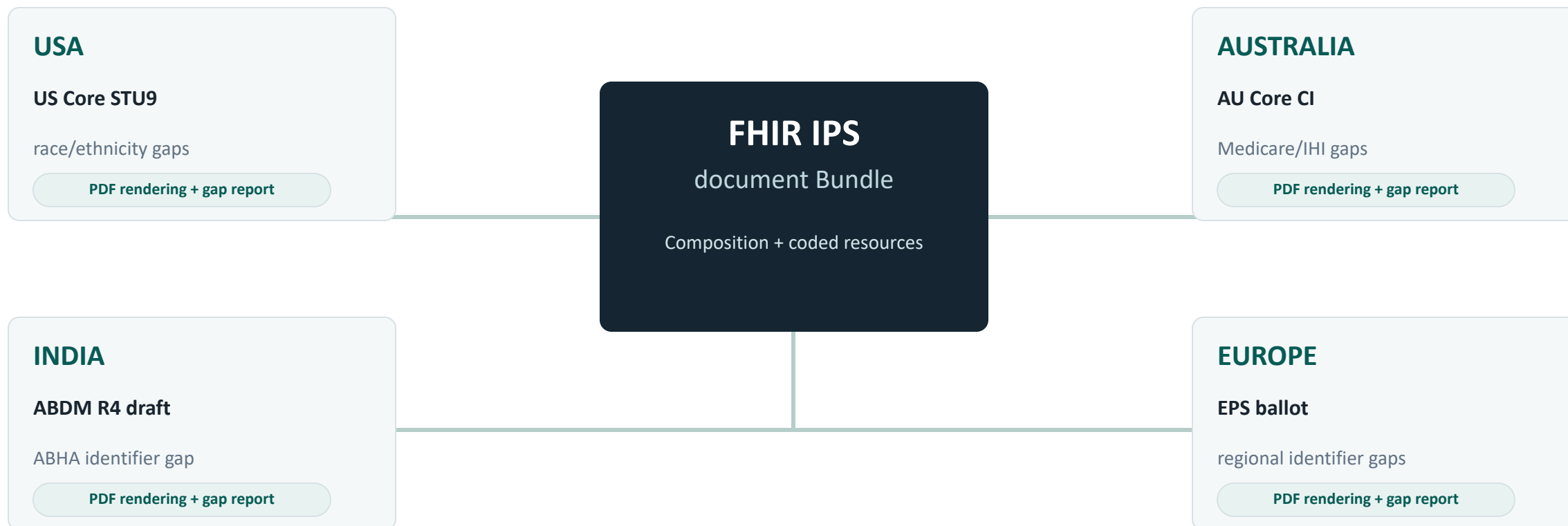
Bundle.type = document

REPRESENTATIVE ROUTE	VALIDATOR	RESULT
USA -> India	HL7 FHIR Validator + IPS 2.0.1	0 errors
India -> USA	HL7 FHIR Validator + IPS 2.0.1	0 errors
Australia -> Europe	HL7 FHIR Validator + IPS 2.0.1	0 errors
Europe -> USA	HL7 FHIR Validator + IPS 2.0.1	0 errors
Claim boundary: validator success is evidence, not production certification or national profile conformance.		

Validator proof is structural and profile-oriented. Semantic correctness is tested separately against synthetic gold labels and trace review.

One IPS Bundle supports four receiver ecosystems

Each receiver gets the same machine-readable IPS, a readable PDF, and an explicit list of readiness gaps.



Readiness checks are informative only: US Core STU9, ABDM R4 draft, AU Core CI, and European Patient Summary ballot materials.

The AI layer ends in HL7 artifacts, evidence, and review boundaries

COMMON PROTOTYPE

OUR JUDGE-VERIFIABLE DIFFERENCE

LLM directly emits FHIR

Registry-first mapping; low confidence stops for review

Mapping hidden in model output

FHIR CodeSystem, ValueSet, ConceptMap, and operation trace

Data pooled centrally

Local training + sample-weighted FedAvg; no raw reports sent

Single destination format

One FHIR IPS artifact + four receiver readiness checks

Structure checked once

Gold-label semantic tests + official validator evidence

REPRODUCIBLE EVIDENCE

29 tests · 48/48 transfer · 0 validator errors · coordinator payload verified · JSON + evidence pack



The interoperable summary travels. The learning stays local.

Federated learning

aligns local clinical phrases without
centralizing raw reports

FHIR terminology

publishes auditable CodeSystem,
ValueSet, ConceptMap, and validation
traces

FHIR IPS

packages coded facts into a machine-
readable cross-border patient
summary

PROPOSED NEXT PILOT

**Two hospitals · 500 governed local mentions per site · independent clinician
adjudication · live terminology server**

Synthetic prototype · readiness checks only · no formal privacy guarantee · no production certification